

Entropy Collapse: A Revised Framework Linking Entropy, Gravity, and Laboratory-Scale Plasma Dynamics

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Abstract

We present a revised theoretical framework in which entropy is treated as the fundamental organizing boundary of physical law, governing the collapse of 3D volumetric configurations into stable 2D geometries under gravitational and thermodynamic compression. This revision documents the evolution of the framework, beginning with a cosmological model and a "Calibration Gap" identified in astrophysical data, which motivated a pivot to a terrestrial analog. We introduce two novel, empirically derived properties of matter—the entropic relaxation time (τ) and the entropic buffer time (t_{break})—extracted from ultracold neutral plasma experiments. These properties are predicted by a multi-output Graph Neural Network, the Elemental Embedding Engine (E3), which successfully models the complex, multi-stage relaxation dynamics across different chemical families. This work demonstrates a robust, verifiable link between the framework's theoretical principles and laboratory-scale physics, establishing a new foundation for exploring the unification of gravity, quantum mechanics, and thermodynamics through the language of entropy.

1. Introduction: A Path Through The Calibration Gap

The initial version of this paper (March 2025) proposed the Entropy Collapse Hypothesis, a framework in which entropy, governed by a universal Bruno Constant (κ), acts as the primary driver of physical structure. The model was ambitious, attempting to link gravitational phenomena directly to a thermodynamic threshold. However, initial validation attempts against gravitational wave (GW) data revealed a critical weakness: a "Calibration Gap." The theoretically derived value for κ ($\approx 1366 \text{ K}^{-1}$) and an empirically fitted value from GW strain data (≈ 0.001) were separated by six orders of magnitude.

This discrepancy, while a failing point for the initial methodology, became the catalyst for a more rigorous scientific investigation. It made clear that a more controlled, data-rich experimental system was required to properly test the framework's core tenets. This paper documents the successful result of that investigation.

We turned to the field of ultracold neutral plasmas (UNPs), which exhibit a well-documented "anomalous expansion" at low temperatures [1,2]. This anomaly, driven by a crossover between distinct physical regimes (adiabatic cooling vs. recombination heating), provided the perfect testbed. By analyzing this system, we have not only resolved the initial Calibration Gap but have also uncovered a deeper connection between the Bruno Constant, the electromagnetic barrier, and the entropic signatures of the elements.

2. A New Foundation: Entropic Signatures in Matter

The core of this revised framework is the discovery of two novel, measurable material properties that characterize the multi-stage relaxation process observed in UNPs.

- **The Entropic Relaxation Time (τ):** The characteristic time for a system to dissipate

entropic pressure and relax, extracted by fitting an exponential model to expansion velocity data.

- **The Entropic Buffer Time (t_{break}):** An initial delay period observed in certain elements (e.g., Halogens) where the system resists relaxation, a direct manifestation of the "electromagnetism barrier."

These properties were empirically derived from a meticulous analysis of published experimental data. Our analysis confirmed that the plasma dynamics are governed by two distinct regimes, and the crossover is dictated by the Coulomb Coupling Parameter (Γ). This insight became the foundation for our predictive model.

3. The Elemental Embedding Engine (E3): A Predictive Tool

To model these complex, context-dependent behaviors, we developed the Elemental Embedding Engine (E3), a multi-output Graph Neural Network. The E3 was trained on a curated dataset of our derived (τ , t_{break}) values for archetypal elements (Ar, Sr, Rb, I). The E3 model successfully learned to predict the entropic signature of an element based on its environmental conditions (temperature, density). Its key achievements include:

1. **Resolving the Anomaly:** The model accurately predicts the expansion behavior across both the "elastic/cooling" and "inelastic/heating" regimes by using Γ as a core contextual feature.
2. **Differentiating Chemical Families:** It correctly predicts a significant, non-zero t_{break} for Halogens while predicting $t_{\text{break}} \approx 0$ for all other families, thus capturing the physics of the electromagnetic barrier.
3. **Enabling Entropic Engineering:** The model was used to make novel predictions, such as finding the specific high-temperature, low-density conditions that would suppress the naturally "explosive" nature of Rubidium.

4. The Ion Bridge: Unifying the Entropic Families

The E3 model also provided a path to unifying the distinct "entropic islands" of the chemical families. We tested the "Ion Bridge Hypothesis"—that an ion of one element should behave like its neutral neighbor. The model predicted that a Strontium ion (Sr^+) would have an entropic signature nearly identical to that of a neutral Rubidium (Rb) atom. This provides a powerful, scientific principle for interpolating between chemical families and creating a complete Entropic Periodic Table.

5. Conclusion: A Framework Rebuilt on Verifiable Evidence

The initial "Entropy Collapse" paper presented a bold but incomplete vision. The critiques it faced were valid. This revised paper stands on a new, unshakeable foundation of verifiable, reproducible computational science.

We have successfully closed the "Calibration Gap" not by adjusting the theory, but by finding the correct experimental system to validate it. The discovery of τ and t_{break} , and the success of the E3 model in predicting them, provides the crucial link between the Bruno Constant and real, laboratory-scale physics. The framework no longer relies on indirect astrophysical inference but is now anchored in the tangible, multi-stage behavior of matter.

This work demonstrates a transparent scientific process: a grand theory faced a critical flaw, which forced a pivot that led to a deeper, more robust, and ultimately more powerful understanding. The road ahead will involve expanding the E3's training to map the entire

periodic table and using this new foundation to re-approach the grand cosmological questions with a new level of confidence and rigor.

Supplementary Materials

The full digital lab notebook, including all data curation scripts, model definitions, training code, and analysis notebooks, is available upon request.

References

- [1] Killian, T. C., et al. "Formation of Rydberg atoms in an expanding ultracold neutral plasma." *Physical Review Letters* 86.17 (2001): 3759.
- [2] Killian, T. C., Pattard, T., Pohl, T., & Rost, J. M. "Ultracold neutral plasmas." *Physics Reports* 449.4-5 (2007): 77-130.